

RESQNET AI

Autonomous, Zero-Network & Voice-First Emergency Response Engine for the Golden Hour

Problem Statement Focus: 1.3 RoadSoS | **Target Institution:** Centre of Excellence for Road Safety (CoERS), IIT Madras

1. Executive Summary & System Vision

In high-impact road accidents, the survival rate depends critically on the "**Golden Hour**"—the first 60 minutes post-trauma. Current solutions fail due to three core systemic barriers: loss of cellular connectivity on remote highways, physical incapacitation of victims preventing active phone interaction, and cognitive panic making manual text inputs impossible.

ResQNet AI re-engineers emergency response by shifting processing from the cloud to the absolute edge. By creating a fully offline architectural pipeline comprising quantized **Small Language Models (SLMs)**, on-device **Vector Databases**, continuous **Sensor Fusion hardware triggers**, and a peer-to-peer **Bluetooth Low Energy (BLE) Mesh recovery layer**, ResQNet AI guarantees reliable life-saving operation even under complete infrastructure collapse.

2. Deep-Dive Technology Stack

To meet the stringent requirements of memory conservation, local execution, and global schema compliance preferred by IIT Madras, the tech stack is entirely open-source, lightweight, and local-first:

Layer	Technology Selected	Architectural Rationale
Core Backend & logic	Python 3.11 / ONNX Runtime Mobile	Optimized execution of compressed machine learning graphs directly inside native device runtimes.
Offline Vector DB	DuckDB + HNSW Vector Extension	Provides lightning-fast analytical SQL and geographical vector index mapping using low memory footprint.
Offline Speech AI	Vosk Audio Engine / Whisper-v3-nano (Quantized)	Local Automatic Speech Recognition (ASR) needing zero internet dependencies, optimized to execute in under 45MB RAM.
Local GIS Engine	OSRM + Highly Compressed OSM (.pbf)	Calculates Haversine/routing distances to the closest trauma centers completely offline without Google Maps APIs.
P2P Mesh Network	BLE (Bluetooth Low Energy) 5.2 / Wi-Fi Direct	Broadcasts localized SOS tokens continuously to surrounding vehicles without cellular reception.

3. Extreme Offline Robustness: Local Data Mapping Architecture

The primary constraint of the RoadSoS problem statement is operational capacity in low-network or zero-connectivity zones. ResQNet AI overcomes this by utilizing a double-layer offline database strategy.

3.1 Local OpenStreetMap (.pbf) Parsing & Spatial Compression

Instead of calling standard cloud-based mapping endpoints, the system bundles regional geographical datasets extracted from OpenStreetMap (OSM) compressed into highly optimized binary chunks. The geospatial infrastructure uses a customized structural schema:

- **Node Pruning:** Filters out non-essential points of interest, keeping only strict categories: Trauma Centers, Emergency Hospitals, Ambulance Base Stations, Towing Services, and Puncture Shops.
- **Geographic Sharding:** The database is divided into administrative grid structures based on coordinates to guarantee predictable execution speeds.

Mathematical Distance Evaluation (Offline Haversine Index):

To instantly rank hospitals based on proximity without a network connection, the local database utilizes the Haversine formula executing compiled vector math over the localized DuckDB tables:

$$d = 2R \cdot \arcsin(\sqrt{(\sin^2(\Delta\phi/2) + \cos(\phi_{\text{sub1}}) \cdot \cos(\phi_{\text{sub2}}) \cdot \sin^2(\Delta\lambda/2))})$$

Where ϕ is latitude, λ is longitude, and R is Earth's radius (6,371 km). This is indexed using DuckDB's spatial bounding structures for sub-millisecond querying.

4. DuckDB & Local Vector Database Engine

A core requirement of an IIT-level entry is ensuring high reliability and precision in data fetching. ResQNet AI moves away from unstructured JSONs, implementing an embedded **DuckDB column store database** integrated with hierarchical navigable small world (HNSW) indices for local semantic similarity matches.

4.1 Query Processing Workflow

When an emergency request is parsed by the local AI, it triggers an optimized query execution pipeline:

1. The system extracts key entities (e.g., entity: "Trauma Center", urgency: "High").
2. DuckDB executes an optimized spatial distance calculation joined against the pre-loaded emergency services index table.
3. The query uses custom structured SQL views that automatically prioritize facilities equipped with advanced triage capabilities based on physical proximity.

Because DuckDB operates as an in-process engine, it avoids inter-process communication overheads, allowing resource-constrained processors to perform complex queries over lakhs of local emergency listings instantly.

5. Hands-Free, Voice-Activated AI Triage Agent

During extreme distress, manual typing introduces unacceptable latency and error rates. ResQNet AI implements a fully localized, voice-driven conversational pipeline.

5.1 Offline Pipeline Processing Sequence

The voice sub-system operates via a continuous audio streaming loop that requires no connection to external cloud engines:

- **On-Device Wake-Word Detection:** A highly optimized local acoustic model listens for explicit distress triggers such as "Help Help", "Accident", or localized regional terms.
- **Quantized Local ASR:** Once triggered, the acoustic signal feeds into an ONNX-quantized Vosk model to generate immediate text transcripts.

- **Deterministic Rule-Based Triage Matrix:** The transcript is evaluated against an localized algorithmic medical framework based on Advanced Trauma Life Support (ATLS) protocols.

Localized Triage Protocol Sample Workflow:

Agent Audio Output: "Emergency detected. Is the patient breathing and conscious?"

User Input: "Conscious but heavy bleeding from the leg."

System Action: Instantly parses keyword "bleeding" → Computes nearest trauma facility → Plays loud, step-by-step audio instructions: *"Apply direct, hard pressure on the wound using a clean cloth immediately. Keep the limb elevated."*

6. Unconsciousness Handling & Autonomous Sensor Fusion

The most critical edge-case of a premium hackathon solution involves handling absolute victim incapacitation (e.g., driver knocked unconscious after a major crash).

6.1 Hardware-Driven Crash Detection Algorithm

ResQNet AI taps directly into the smartphone's integrated hardware sensors (Three-axis Accelerometer and Gyroscope) through a continuous high-frequency polling loop:

- **Impact Threshold Verification:** The app monitors mathematical Vector Magnitude Accelerations (A_{mag}). If a sudden change exceeding predetermined structural impact thresholds (e.g., $A_{mag} \geq 4.5G$ deceleration) occurs concurrently with a violent angular velocity change via the gyroscope, a high-priority crash event is declared.
- **Fail-Safe Triage Intercept:** The application immediately sounds an intensive auditory alarm countdown for exactly 10 seconds. If the user does not manually override or tap to cancel, the system assumes complete loss of consciousness and switches immediately into **Autonomous Emergency Mode**.

6.2 Infrastructure Collapse Fallback: BLE Disaster Mesh Networks

If the unconscious crash occurs in a deep dead-zone with absolute cellular network failure, ResQNet AI launches its ultimate safety subsystem: **Peer-to-Peer Disaster Mesh Beaconing**.

The device instantly configures its hardware transmitter into a continuous loop broadcasting an encrypted **Asynchronous BLE Disaster Token**. This token encodes critical localized telemetry packed into a highly compressed byte string:

```
[Token Structure: TIMESTAMP (4B) | LAT (4B) | LON (4B) | BLOOD_TYPE (1B) |  
                    IMPACT_SEVERITY (1B)]
```

Any vehicle passing within a 150-meter radius running the ResQNet background service will transparently capture this token via BLE scanning. The passing phone stores this token in its local queue. The moment that passing vehicle moves into an area with cellular connectivity, its app automatically uploads the token data to the central emergency cloud backend, relaying the exact GPS coordinate coordinates of the unconscious victim to dispatch centers.

7. Global Applicability & Standardized Data Layer

To fulfill the mandatory requirement of "**Global applicability across countries**" highlighted in the IIT Madras guidelines, ResQNet AI decouples its execution logic from regional data schemas.

7.1 Mobile Country Code (MCC) Autonomous Routing Engine

The software dynamically adapts its emergency execution environment on launch without requesting network configuration protocols:

1. The system reads the hardware SIM card's native **Mobile Country Code (MCC)** directly via baseband APIs.
2. If the MCC matches India (**404/405**), the routing engine maps emergency targets dynamically to 112 (National Emergency) and 108 (Ambulance), pulling target DuckDB datasets mapped to Indian states.
3. If the user travels internationally and the system reads an MCC corresponding to the United States (**310/311**), the engine instantly maps its core dialing logic to 911, and switches geospatial profiles to match North American trauma levels seamlessly.

By utilizing a universal, standardized global schema layer based on international geographic data standards, a single compiled build can be deployed globally without code refactoring.

8. Execution Feasibility & Strategic Roadmap

ResQNet AI is engineered not just as a concept, but as a production-grade software system designed to integrate natively into existing government frameworks.

- **Government Partnerships:** Designed to plug into the Ministry of Road Transport and Highways (MoRTH) and NHAI database servers during regular sync intervals to fetch updated lists of accredited trauma facilities.
- **Low Integration Friction:** The complete edge architecture ensures that the system costs zero overhead in active server compute, scaling linearly across millions of vehicles without requiring expensive cloud infrastructure backend expansion.